

Class Project U1 - U2. Executive EDA Report

Guidelines

The purpose of this project is to conduct an Exploratory Data Analysis using real-world datasets and communicate the most relevant findings through an Executive EDA Report. Students will pick one project at random from six available options:

1. WHO Triple Billion Project: <https://data.who.int/dashboards/global-progress/triple-billion>
2. WHO Health data overview for Mexico: <https://data.who.int/countries/484>
 - a. Population
 - b. Life expectancy
 - c. Leading causes of death
 - d. Health statistics
3. WHO Health data overview for USA: <https://data.who.int/countries/840>
 - a. Population
 - b. Life expectancy
 - c. Leading causes of death
 - d. Health statistics
4. WHO Health data overview for Canada: <https://data.who.int/countries/124>
 - a. Population
 - b. Life expectancy
 - c. Leading causes of death
 - d. Health statistics
5. INEGI Population and Housing Censuses and Counts 2020: <https://www.inegi.org.mx/datosabiertos/>
 - a. Key results by location
 - b. Main results by AGEb and urban block
 - c. Census statistics at geo-electoral scales
6. INEGI Surveys: <https://www.inegi.org.mx/datosabiertos/>
 - a. Regular Surveys

Content

The Executive EDA Report must contain

- Title: A clear and specific title related to the data being analyzed.
- Infographic: One-page infographic that provides a visual overview of the entire analysis. (1 page)
- Executive Summary: It must provide a clear explanation of the dataset, the context of the problem, and the relevance of the analysis. In general, it should explain things like what data were analyzed? Where do the data come from? What are the main variables included in the analysis? Why might the analysis be relevant for decision-making? etc. (1 to 2 pages)
- Exploratory Data Analysis: Students must explore the dataset using appropriate statistical and graphical techniques. There is no predetermined number of plots that must be included. Students must decide which analyses and visualizations are the most useful for understanding the dataset and observe whether the data exhibit characteristics such as trend, seasonality, cyclical behavior, outliers, different patterns across time. (3 to 5 pages).



- **Conclusions:** This part should summarize the most important knowledge obtained from the Exploratory Data Analysis. Students could resume the description of plots and figures. Also integrate the most important findings and explain what has been learned from the dataset (1 page).
- **Appendices:** Provide the original dataset used in the project and include the complete Python script used to perform the analysis (1 page).

The final report must be sent via email before the deadline designated by the professor.